

Depth-Conditioned Semantic Communication for Stereo 3D Object Detection

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Abstract—Wireless stereo 3D detection requires a communication representation that retains both object appearance and depth-dependent matching information. Reconstructing the stereo images introduces an intermediate objective whose image-quality ranking can differ from the detection ranking. We propose a conditional depth-field representation for transmitting stereo features to a fixed 3D detector. At each spatial location, the encoder divides a learned depth distribution into three equal-mass intervals and transmits their barycenters together with conditional feature vectors. A compact appearance code completes the packet, requiring 19 complex channel uses per location. The receiver reconstructs the cost field at metric depth queries from the received coordinates and features. We further formulate a receiver that uses channel state information and node ordering to average the reconstruction basis over a finite-grid posterior. Experiments on 3,769 KITTI stereo pairs show that higher RGB PSNR can coincide with lower 3D average precision across source codecs. In separate simulations under a specified ordered prior, posterior decoding reduces latent-coordinate mean squared error from 69.3591 to 26.5377 m² over a Rayleigh channel at 10 dB. These results characterize the motivation and receiver mechanism; the proposed representation’s end-to-end detection gain remains to be evaluated.

Index Terms—Task-oriented communication, stereo vision, 3D object detection, joint source-channel coding, depth-conditioned features.

I. INTRODUCTION

A wireless stereo perception system must convey enough visual information for a remote detector to recognize objects and estimate their 3D locations. These two requirements place different demands on the transmitted representation. Appearance supports recognition, whereas left–right correspondence determines depth. Compressing a stereo pair to improve an image distortion measure therefore need not preserve the information most useful to the detector. The communication problem is to allocate a limited channel budget to the features that support the final detection decision.

Learned joint source-channel coding provides a way to optimize a transmitted representation through a differentiable channel model. For example, Yilmaz et al. [1] jointly encode and reconstruct images sent over a multiple access channel. Task-oriented systems instead select features for the intended inference task. Huang et al. [2] use task- and SNR-dependent feature extraction in a multi-task semantic communication system. These designs motivate optimizing the communication interface jointly with its reconstruction or inference objective. For stereo 3D perception, the remaining design choice is how to represent the depth structure on which the task depends.

Cao et al. [3] address this problem by communicating optical-flow and object-region information and reconstructing stereo images at the receiver. Their image interface can be followed by an RGB-based stereo detector. Stereo R-CNN [4] performs stereo object association and region-based alignment, while LIGA-Stereo [5] learns stereo representations with LiDAR geometry guidance during detector training. An alternative communication interface can be placed inside such a detector: a sender extracts stereo features, and a receiver performs detection from the transmitted representation. Feature communication is also studied in collaborative perception, including digital compression and reliability-aware feature selection in CoDS [6]. The question here is more specific: how should a small packet represent the joint variation of stereo features and metric depth?

A dense bottleneck compresses the depth axis into a fixed set of latent channels. We consider a representation whose transmitted coordinates explicitly specify where features belong along that axis. At each pooled spatial location, three depth barycenters summarize a learned depth weighting, and each barycenter carries a feature conditioned on its depth interval. The receiver uses these coordinate–feature pairs to reconstruct the cost field required by a frozen detector. This design creates a direct connection between the packet and its task interface: coordinate errors move feature support along the depth axis, while feature errors change its content. It also permits the receiver to account for uncertainty in the received coordinates when evaluating the reconstruction basis.

This paper develops the following contributions:

- A conditional depth-field codec that represents each spatial location by three equal-mass depth nodes, their associated features, and an appearance code. The complete packet has a fixed budget of 19 complex symbols and energy 19 per location.
- A channel-conditioned reconstruction rule that combines the likelihood of the received position symbols with an ordered-node prior. Finite-grid sum-product inference provides the node marginals, which are used to average the field basis rather than evaluate it at a single coordinate.
- An empirical analysis of two design questions: whether RGB quality ranks stereo detection quality consistently, and how coordinate uncertainty affects field reconstruction. Full-split source-codec results and synthetic channel results address these questions; matched-budget detection comparisons are specified separately.

II. SYSTEM MODEL AND DESIGN OBJECTIVE

A. Stereo perception interface

We consider a calibrated stereo camera pair connected to a remote 3D detector through a wireless link. A frozen stereo front end processes the left and right images and produces cost features $\mathbf{c}_{sd} \in \mathbb{R}^{32}$, appearance features $\mathbf{a}_s \in \mathbb{R}^{32}$, and a depth guide π_{sd} . The index s denotes a spatial location and d a depth query. We use $D = 72$ metric queries,

$$z_d = 2 + \frac{57.6}{71}d \text{ m}, \quad d = 0, \dots, 71. \quad (1)$$

These queries follow the detector's interpolation interface; the internal cost-volume sweep has its own coordinate convention. Spatial average pooling with stride four reduces all three tensors, and the pooled guide is normalized across depth. The guide is a network prediction used to weight features, rather than an assumed calibrated depth posterior.

The role of the depth axis follows from stereo geometry. For focal length f , camera baseline B , and disparity d_{disp} ,

$$Z = \frac{fB}{d_{\text{disp}}}, \quad \Delta Z \simeq -\frac{fB}{d_{\text{disp}}^2} \Delta d_{\text{disp}}. \quad (2)$$

Thus a fixed disparity perturbation has a larger first-order effect at small disparities. This motivates retaining depth-dependent feature structure. Detection performance additionally depends on recognition and box localization, so we optimize the codec using the detection loss.

B. Wireless channel and resource constraint

The transmitted packet $\mathbf{x} \in \mathbb{C}^N$ contains N complex channel uses and satisfies $\|\mathbf{x}\|_2^2 = N$. For each symbol,

$$y = hx + n, \quad n \sim \mathcal{CN}(0, q), \quad q = 10^{-\rho/10}, \quad (3)$$

where ρ is the SNR in dB. We consider AWGN with $h = 1$ and independent Rayleigh fading with $h \sim \mathcal{CN}(0, 1)$ for each symbol. Under perfect receiver channel state information (CSI), zero-forcing equalization gives $r = y/h$ with conditional noise variance $q/|h|^2$. Fades are neither clipped nor replaced by an erasure rule. The model assumes ideal CSI and does not include pilot or channel-estimation overhead.

Let E_θ and R_θ denote the communication encoder and decoder, F_0 the frozen stereo front end, and H_0 the frozen detection head. For a stereo input $\mathbf{I}$ with detection targets $\mathbf{t}$, the training objective is

$$\min_{\theta} \mathbb{E}[\mathcal{L}_{\text{det}}(\hat{\mathbf{t}}, \mathbf{t})], \quad (4)$$

subject to the packet length and energy constraint. The expectation is over training samples and channel realizations, and $\mathcal{H}$ denotes the channel and equalization. Only the codec is optimized. The receiver knows the shared model parameters and tensor dimensions; all sample-dependent features and coordinates must pass through the channel.

III. CONDITIONAL DEPTH-FIELD COMMUNICATION

A. Depth-conditioned feature encoding

At location s , a pointwise network g_θ with dimensions $32 \rightarrow 16 \rightarrow 1$ combines the cost feature with the native depth guide to produce normalized weights:

$$p_{sd} = \text{softmax}_d(g_\theta(\mathbf{c}_{sd}) + \log \pi_{sd}). \quad (5)$$

The guide is floored at the minimum positive floating-point value before taking the logarithm. This weighting lets the codec use both the pretrained stereo guide and the features selected by the detection objective.

To obtain a fixed-size representation, we partition the cumulative depth mass into $K = 3$ intervals. Define $F_{sd} = \sum_{t \leq d} p_{st}$ and $F_{s,-1} = 0$. The mass contributed by depth atom d to interval j is

$$m_{sjd} = \left[\min \left(F_{sd}, \frac{j+1}{K} \right) - \max \left(F_{s,d-1}, \frac{j}{K} \right) \right]_+, \quad (6)$$

where $j = 0, 1, 2$ and $[v]_+ = \max(v, 0)$. With $w_{sjd} = m_{sjd} / \sum_t m_{sjt}$, the transmitted node and its conditional feature are

$$\mu_{sj} = \sum_d w_{sjd} z_d, \quad \mathbf{v}_{sj} = \sum_d w_{sjd} A_\theta(\mathbf{c}_{sd}), \quad (7)$$

where $A_\theta : \mathbb{R}^{32} \rightarrow \mathbb{R}^8$ is a learned pointwise map. Each node summarizes one third of the depth mass in exact arithmetic, and the interval construction orders the barycenters. It retains three conditional summaries rather than collapsing all depth support into one mean. This is a representation choice; its suitability for detection is tested against a dense depth bottleneck in Section IV.

B. Channel mapping and field reconstruction

We encode each barycenter as a unit-energy phase symbol,

$$x_{sj} = e^{i\vartheta_{sj}}, \quad \vartheta_{sj} = \pi \frac{\mu_{sj} - 2}{57.6} - \frac{\pi}{2}. \quad (8)$$

Each eight-dimensional real feature is paired into four complex symbols and normalized jointly to energy four. A separate $32 \rightarrow 8$ appearance encoder produces another four-symbol group with energy four. The packet therefore contains

$$N = N_{\text{sites}} \underbrace{(3 + 3 \cdot 4 + 4)}_{\text{positions + features + appearance}} = 19N_{\text{sites}} \quad (9)$$

complex symbols, with the same total energy. The fixed detector interface has 1,560 pooled locations, giving 29,640 complex uses per stereo frame. Normalization retains feature direction; discarded magnitudes are not sent as side information. Near-zero feature groups use a fixed sphere point.

For the primary receiver, the equalized position symbols are projected onto the unit circle. Their phases are clipped to $[-\pi/2, \pi/2]$ and mapped back to coordinates $\hat{\mu}_{sj} \in [2, 59.6]$ m. Each received feature group is projected onto its

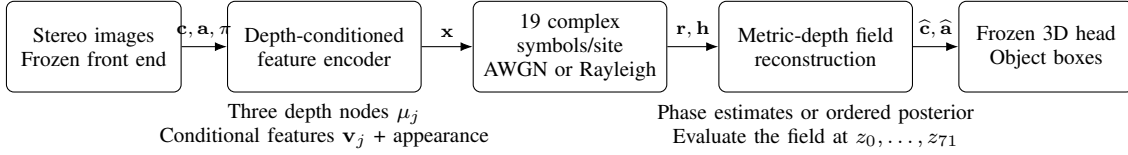


Fig. 1. Conditional depth-field communication. The encoder transmits depth coordinates and associated features to reconstruct the detector’s cost field. Training updates the codec through the frozen detection head. The ordered posterior is a receiver extension; the primary codec comparison uses phase-based reconstruction.

known energy-four sphere and decoded with $D_\theta : \mathbb{R}^8 \rightarrow \mathbb{R}^{32}$. The reconstructed cost field is

$$\hat{\mathbf{c}}_{sd} = \frac{1}{3} \sum_{j=0}^2 \exp \left[-\frac{(z_d - \hat{\mu}_{sj})^2}{2\sigma^2} \right] D_\theta(\hat{\mathbf{v}}_{sj}), \quad (10)$$

where $\sigma > 0$ is a learned scalar shared across all locations. The Gaussian basis places each decoded feature around its received depth coordinate. Its weights are not normalized across nodes. An independent $8 \rightarrow 32$ map decodes appearance. Trilinear interpolation restores the cost tensor and bilinear interpolation restores the appearance tensor to the shapes consumed by the frozen detection head.

C. Reconstruction under coordinate uncertainty

A point estimate in (10) treats an uncertain position symbol as one known coordinate. Under fading, equalized observations can have very different reliabilities. We therefore consider a receiver extension that averages the basis using the coordinate likelihood and node ordering. This extension changes neither the transmitted packet nor its energy.

Suppress the spatial index and write $u_j = (\mu_j - 2)/57.6$. On a shared grid $\{u_m\}_{m=1}^M \subset [0, 1]$, the position log-likelihood, up to terms independent of m , is

$$\ell_j(m) = \frac{2|h_j|^2}{q} \Re \left\{ r_j e^{-i\pi(u_m - 1/2)} \right\}. \quad (11)$$

The factor $|h_j|^2/q$ accounts for the actual reliability of the equalized observation. We use a uniform prior over ordered grid triples $a \leq b \leq c$, including ties. For $L_j(m) = e^{\ell_j(m)}$, the forward messages are

$$\begin{aligned} A_1(m) &= L_1(m), \\ A_2(m) &= L_2(m) \sum_{t \leq m} A_1(t), \\ A_3(m) &= L_3(m) \sum_{t \leq m} A_2(t). \end{aligned} \quad (12)$$

The backward messages satisfy $B_3(m) = 1$, $B_2(m) = \sum_{t \geq m} L_3(t)$, and $B_1(m) = \sum_{t \geq m} L_2(t) B_2(t)$. The marginal for node j is $A_j(m) B_j(m) / \sum_t A_3(t)$. Log-domain prefix and suffix sums compute these marginals in $O(3M)$ operations using standard sum-product inference.

The posterior-averaged basis weight is

$$\bar{k}_{jd} = \frac{1}{3} \sum_m p(u_j = u_m \mid \mathbf{r}, \mathbf{h}) \exp \left[-\frac{(z_d - 2 - 57.6u_m)^2}{2\sigma^2} \right]. \quad (13)$$

The receiver substitutes these weights into (10) while keeping the feature and appearance decoders fixed. Because the basis is nonlinear, evaluating it at the posterior-mean coordinate generally differs from averaging its values. The latter estimates the basis under the assumed posterior and is the quantity examined in the synthetic field experiment. The model omits feature-coordinate dependencies, and its uniform ordered prior is not fitted to the learned KITTI coordinates. We use 289 grid points for receiver integration; the synthetic validation in Section IV-C uses a separate 33-point source prior.

D. Task-driven codec training

We optimize the codec with the detector’s native classification and box-regression loss, including localization, IoU, and direction terms. The stereo front end and detection head remain frozen throughout training. LIGA-Stereo’s original pretraining uses LiDAR guidance [5]; codec training uses stereo images and detection labels without an additional RGB reconstruction or depth-supervision loss.

Each codec is trained on 3,712 stereo pairs for three epochs using Adam, learning rate 10^{-3} , and zero weight decay. Training samples AWGN and Rayleigh channels with equal probability and draws SNR continuously from [6, 18] dB. Seed 17 fixes the frame order and channel schedule. Compared codecs use paired noise and fading realizations. This setup optimizes the communication representation for the same fixed task receiver and channel range.

IV. EXPERIMENTAL RESULTS

A. Evaluation setup and comparisons

Evaluation uses all 3,769 pairs of the fixed KITTI validation split. The detection metric is Car 3D Moderate AP_{R40} at IoU 0.7. The primary comparison uses the third-epoch checkpoint at AWGN 10 dB, without selecting checkpoints by validation AP. The planned channel sweep covers AWGN at integer SNRs from 6 to 18 dB and Rayleigh at 6, 8, 10, 12, 14, 16, and 18 dB, together with an identity-channel reference.

Table I defines the four codecs. P denotes the proposed guide-conditioned field codec. G removes the guide term from (5), while S circularly shifts the guide by half the pooled height and width. These controls test guide usage and its spatial correspondence. B is a dense codec that concatenates the centered log guide with the cost features, applies a $33 \rightarrow 10$ feature map and learned $72 \rightarrow 3 \rightarrow 72$ depth maps, and reconstructs the cost channels with a $10 \rightarrow 32$ map. Its

TABLE I
CODEC COMPARISONS AT A COMMON BUDGET OF 19 COMPLEX SYMBOLS
AND ENERGY 19 PER SPATIAL LOCATION. PARAMETER COUNTS EXCLUDE
THE FROZEN DETECTOR.

Codec	Representation	Guide	Positions	Params.
G	Conditional field	None	3	1,650
P	Conditional field	Aligned	3	1,650
S	Conditional field	Shifted	3	1,650
B	Dense bottleneck	Aligned	0	1,751

TABLE II
END-TO-END DETECTION COMPARISON TO BE COMPLETED.
THIRD-EPOCH CHECKPOINTS; 3,769 VALIDATION PAIRS; AWGN 10 dB;
CAR 3D MODERATE AP_{R40} AT IOU 0.7.

Codec	G	P	S	B
AP (%)	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>

15 complex feature symbols form one energy-15 group; the appearance interface is shared with the field codecs.

P-B compares complete architectures with the same channel budget, including their different parameter counts and feature normalization groups. P-G and P-S isolate changes within the field architecture. The primary comparison uses phase-based decoding; the posterior extension is evaluated separately. Full-split learned-codec AP is pending in Table II. The completed results below concern source compression and synthetic receiver estimation. The single training seed does not quantify training variability.

B. Image quality and detection quality

We first examine whether image quality consistently ranks stereo detection performance. JPEG and JPEG2000 reconstructions at nominal compression ratios (CRs) of 10, 30, and 50 are evaluated with frozen Stereo R-CNN and LIGA-Stereo detectors. These experiments cover the full validation split and contain no wireless channel. Actual CR is the total raw RGB byte count divided by the total compressed byte count. PSNR is calculated from pooled full-image RGB MSE.

Table III shows a ranking discrepancy at nominal CRs 10 and 30. At CR 10, JPEG2000 yields 36.093 dB PSNR, compared with 29.915 dB for JPEG, yet its LIGA AP is 62.650%, compared with 65.019% for JPEG. Stereo R-CNN exhibits the same direction of change. At CR 50, JPEG2000 improves both PSNR and AP. Thus PSNR does not consistently order downstream detection quality across these operating points. This observation motivates a task loss and a representation tied to the detector’s depth features. The unequal source rates and different codec artifacts do not identify the cause of the discrepancy or establish a matched-rate advantage.

C. Coordinate and field reconstruction over the channel

To examine the receiver independently of detector training, we sample 65,536 triples per channel condition from a uniform prior over all 6,545 inclusive ordered triples on a 33-point grid. Each triple is transmitted using three unit-energy position

symbols. The 20 AWGN and Rayleigh conditions use the SNR grids specified above, giving 1,310,720 trials. The source prior matches the posterior model in this experiment.

Table IV reports the coordinate errors at 10 dB. Under Rayleigh fading, conditioning on the actual CSI reduces MSE from 31.5355 for the fading-blind ordered receiver to 26.5377. The two receivers coincide under AWGN, where all symbols have the same channel gain. This comparison illustrates the value of weighting coordinate observations by their conditional reliability. The independent uniform-slot posterior has different marginal priors from the ordered model, so its comparison with the ordered receiver combines marginal-prior and inter-node effects. Isolating the ordering effect requires a matched-marginal control.

We also reconstruct a scalar field using fixed coefficients (0.8, −0.3, 0.5), nine query locations, and bandwidth 1/3 in normalized coordinates. Posterior basis averaging gives field MSEs of 0.00059813 under AWGN and 0.00110864 under Rayleigh. Substituting posterior-mean coordinates into the basis gives 0.00060495 and 0.00116890, respectively. These results are consistent with estimating the nonlinear basis directly under the specified posterior. They quantify a receiver mechanism under its assumed source model; transferring this gain to trained stereo features requires end-to-end detection evaluation.

D. Scope of the current evidence

The proposed interface requires the sender to compute stereo cost features and the receiver to use the corresponding detection head. Its small codec parameter count therefore does not measure total sender complexity. The current model also uses a fixed symbol budget, a global basis bandwidth, and perfect CSI. The posterior extension introduces a grid prior and additional computation whose effect on detector accuracy and latency remains unmeasured.

A complete wireless comparison must express digital baselines in channel uses after quantization, framing, channel coding, modulation, and required side information. Source byte ratios alone cannot be compared with the analog budget in (9). Flow/ROI, SRCNN, and KITTI-adapted ECSIC baselines, full detection curves, and sender/receiver resource measurements remain to be completed. The available evidence supports the image/task ranking observation and the synthetic receiver analysis, while leaving the central matched-budget detection comparison open.

V. CONCLUSION

We proposed a semantic communication interface that represents a stereo cost field through depth coordinates and conditional features. Equal-mass depth summaries give a fixed 19-symbol packet per location, and the receiver reconstructs the field at the detector’s metric queries. An ordered posterior extends this reconstruction to account for channel-dependent coordinate uncertainty. Completed experiments reveal an RGB-quality/detection ranking discrepancy and lower synthetic estimation error under the stated prior. Full-split

TABLE III

SOURCE-COMPRESSION RESULTS ON 3,769 KITTI STEREO PAIRS WITHOUT CHANNEL NOISE. AP IS CAR 3D MODERATE AP_{R40} AT IOU 0.7 (%). ACTUAL SOURCE RATES DIFFER AT EACH NOMINAL SETTING, AS REPORTED BY CR AND MEAN BYTES PER STEREO PAIR.

Source codec	Nominal CR	Actual CR	Bytes/pair	PSNR (dB)	Stereo R-CNN AP	LIGA AP
JPEG	10	11.019	252,943	29.915	33.883	65.019
JPEG2000	10	10.004	278,596	36.093	32.188	62.650
JPEG	30	33.205	83,937	27.993	29.947	56.954
JPEG2000	30	30.018	92,849	30.438	28.557	51.681
JPEG	50	54.666	50,985	26.803	20.797	44.571
JPEG2000	50	50.037	55,702	28.652	25.228	45.458

TABLE IV

SYNTHETIC COORDINATE MSE AT 10 dB UNDER THE ORDERED 33-POINT PRIOR. VALUES ARE SCALED BY 57.6^2 INTO M^2 AND DESCRIBE TRANSMITTED LATENT COORDINATES, NOT DETECTED OBJECT DEPTHS.

Receiver	AWGN	Rayleigh
Clipped phase	16.1234	69.3591
Independent uniform-slot posterior	15.4297	46.4875
Weighted isotonic projection	14.3532	40.8337
Ordered fading-blind posterior	12.9310	31.5355
Ordered CSI-conditioned posterior	12.9310	26.5377

wireless detection and resource comparisons are needed to determine whether the representation improves the accuracy-communication tradeoff for stereo 3D perception.

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